Solving polynomial equations

Note: The following section is largely motivational. It attempts to motivate the use of matrices with entries in various domains and fields to solve equations. Students are encouraged to read through and try the exercises. However, we will not directly examine questions about this material in examinations.

We will examine algebraic approaches to solving polynomial equations.

To begin with, we note a property of polynomial rings which accounts for their ubiquity.

Exercise: Given a ring R (not necessarily commutative) with identity show that there is a natural ring homomorphism $\mathbb{Z} \to R$ that sends 1 to the identity in R.

When this homomorphism is one-to-one we can think of $\mathbb{Z}$ as a subring of R.

If R is a domain in the sense that the product of non-zero elements is non-zero, then the kernel of this homomorphism is of the form $p\mathbb{Z}$ for some prime number p. In this case, we can think of the field $\mathbb{F}_p = \mathbb{Z}/\langle p \rangle$ as a subring of R. In this case, R is a vector space over $\mathbb{F}_p$.

More generally, suppose we are given a *commutative* Z and a ring homomorphism $Z \to R$ which sends 1 to the identity in R.

Exercise: Giving a commuting set $\{r_1, \ldots, r_n\}$ of elements of R is the same as giving a ring homomorphism $Z[X_1, \ldots, X_n] \to R$. (Hint: The element r_i of R is the image of X_i.)

When $Z \to R$ is one-to-one we often write the image of this ring homomorphism as $Z[r_1, \ldots, r_n]$. Note that this is a commutative ring even if R is not a commutative ring. (This notation can be confusing when used without the assumption that $Z \to R$ is one-to-one.)

When the above homomorphism is surjective (onto) we say that R is *generated* by $r_1, \ldots, r_n$ over Z.

If there is some (unspecified) collection of elements of R for which this is true, we say that R is *finitely generated over Z*.

When there is a surjective (onto) homomorphism $\mathbb{Z}[X_1, \ldots, X_n] \to R$ for some n, we say that R is *finitely generated* as a ring.

In both the above cases, we see that R is a commutative ring.

Given a subset S of $\mathbb{Z}[X_1, \ldots, X_r]$ of polynomials in finitely many variables $X_1, \ldots, X_n$, we have the quotient ring $R = \mathbb{Z}[X_1, \ldots, X_n]/I$ where $I = \langle S \rangle$ is the ideal generated by S. The algebraic study of the subset S of polynomials can be related to the study of R.

Thus, we see that in number theory, we are primarily interested in finitely generated rings. We will also study rings that are finitely generated over some domains (and fields) that arise along the way.

In this section, we will examine some basic structure results for such rings.

Companion Matrix

Given a monic polynomial $\mathbf{a}$ of degree d in $Z[X]$ where Z is a domain; this means that a_d is a unit in Z and $a_k = 0$ for $k > d$.

Long division allows us to write any polynomial $\mathbf{b}$ in $Z[X]$ in the form

$$\mathbf{b} = \mathbf{q} \cdot \mathbf{a} + \mathbf{r}$$

where $\mathbf{r}$ of degree at most $d - 1$. In analogy with integers, we think of $\mathbf{r}$ as $\mathbf{b}$ *modulo* $\mathbf{a}$. Note that $\mathbf{r}$ is uniquely determined by $\mathbf{b}$ and the monic polynomial $\mathbf{a}$.

Thus, the quotient ring $Z[X]/\langle \mathbf{a} \rangle$ can be identified with the set $Z \oplus ZX \oplus \cdots \oplus ZX^{d-1}$ of polynomials of degree at most $e - 1$. On this set, we can define addition and multiplication via polynomial multiplication *modulo* $\mathbf{a}$ as usual.

Multiplication by X on $Z[X]/\langle \mathbf{a} \rangle$ can be expressed in the *basis* $1, X, \ldots, X^{d-1}$ using the $d \times d$ square matrix

$$C_{\mathbf{a}} = \begin{pmatrix} 0 & 0 & \ldots & 0 & -a_0/a_d \\ 1 & 0 & \ldots & 0 & -a_1/a_d \\ 0 & 1 & \ldots & 0 & -a_2/a_d \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & 1 & -a_{d-1}/a_d \end{pmatrix}$$

with entries in Z. This is called the companion matrix of $\mathbf{b}$.

Exercise: Check that the map that sends X to $C_{\mathbf{a}}$ gives an isomorphism of $Z[X]/\langle \mathbf{a} \rangle$ with a subring of the ring $M_d(Z)$ of $d \times d$ square matrices with entries in Z.

In summary, given a monic polynomial $\mathbf{a}$ of degree d with coefficients in Z, the ring $R = Z[X]/\langle \mathbf{a} \rangle$ has the following structure:

- $R = Z \oplus Z \cdot X \oplus \cdots \oplus Z \cdot X^{d-1}$.
- Multiplication by X on R is given by a $d \times d$ matrix $C_{\mathbf{a}}$ with entries in Z.
- R is identified with the subring $Z[C_{\mathbf{a}}]$ of the ring $M_d(Z)$ of $d \times d$ matrices with entries in Z.
- The kernel of the homomorphism $Z[X] \to M_d(Z)$ that sends X to $C_{\mathbf{a}}$ is generated by $\mathbf{a}$.

We say that $\mathbf{a}$ is the *minimal polynomial* of $C_{\mathbf{a}}$.

Informally, we can say that the companion matrix is a "root" of the polynomial $\mathbf{a}$ in the ring of $d \times d$ matrices over Z.

Exercise: Write the companion matrix W for the polynomial $X^2 - X + 1$ and check that $W^3 = I_2$, the 2×2 identity matrix.

Non-monic case. Given a non-constant polynomial $\mathbf{a}$ in $Z[X]$ of degree d, we have $c = a_d \neq 0$.

Consider the subring $Z_c = Z[1/c]$ consisting of elements of the form a/c^n in the field $Q(Z)$ of fractions of Z. Note that Z_c is a domain which contains Z. Moreover, $\mathbf{a}$ is a monic polynomial in $Z_c[X]$ and

$$C_{\mathbf{a}} = \begin{pmatrix} 0 & 0 & \dots & 0 & -a_0/a_d \\ 1 & 0 & \dots & 0 & -a_1/a_d \\ 0 & 1 & \dots & 0 & -a_2/a_d \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -a_{d-1}/a_d \end{pmatrix}$$

has entries in Z_c.

Thus, $C_{\mathbf{a}}$ is a "root" of $\mathbf{a}$ in the subring $Z_c[C_{\mathbf{a}}]$ of $M_d(Z_c)$.

In other words, given a non-zero polynomial $\mathbf{a}$ of positive degree d, we can study the ring $R = Z[X]/\langle \mathbf{a} \rangle$ with the ring R_c where c is a non-zero element of Z:

- $R_c = Z_c \oplus Z_c \cdot X \oplus \cdots \oplus Z_c \cdot X^{d-1}$.
- Multiplication by X on R_c is given by a $d \times d$ matrix $C_{\mathbf{a}}$ with entries in Z_c.
- R_c is identified with the subring $Z_c[C_{\mathbf{a}}]$ of the ring $M_d(Z_c)$ of $d \times d$ matrices with entries in Z_c.
- The kernel of the homomorphism $Z_c[X] \to M_d(Z_c)$ that sends X to $C_{\mathbf{a}}$ is generated by $\mathbf{a}$.

This description does not give use the structure of R, but is still useful.

Multi-variable case. Given a non-constant polynomial $\backepsilon$ in $Z[X_1, \dots, X_n]$, we can generalise the above construction using the following exercises.

Exercise: Given a tuple of non-negative integers $k_1, \dots, k_n$ and $N > k_i$ for all i, the integers $d = \sum_{i=1}^{n} k_i N^{n-i}$ and N uniquely determine the ordered tuple $(k_1, \dots, k_n)$. (Hint: Note that this is precisely the expression of d in base N.)

We apply this to monomials using the following exercise:

Exercise: Given a monomial $\prod_{i=1}^{n} X_i^{k_i}$ and $N > k_i$ for all i, the leading term of the polynomial $X_n^{k_n} \cdot \prod_{i=1}^{n-1} \left(X_i + X_n^{N^{n-i}} \right)^{k_i}$ is X_n^d where $d = \sum_{i=1}^{n} k_i N^{n-i}$. (Hint: Prove this by induction on n. Also note that any term involving X_i for $i < n$ has strictly smaller degree.)

We can apply this to the polynomial ring by using the following exercise:

Exercise: Given any integer $N > 1$, the ring endomorphism ψ of $\mathbb{Z}[X_1, \dots, X_n]$ given by $\psi(X_n) = X_n$ and $\psi(X_i) = X_i + X_n^{N^{n-i}}$ for $i < n$ is an automorphism of this ring. (Hint: Check that η defined by $\eta(X_n) = X_n$ and $\eta(X_i) = X_i - X_n^{N^{n-i}}$ is the inverse ring endomorphism.)

We now apply the above exercises by using an N that is larger than any power of any X_i that occurs in the finitely terms that occur in $\mathbf{a}$.

Exercise: There is an automorphism ψ of the polynomial ring (which can be seen as a "change of co-ordinates") such that $\psi(\mathbf{a})$ when seen as a polynomial in X_n with coefficients in $Z[X_1, \ldots, X_{n-1}]$ has leading degree term of the form cX_n^d for a suitable c in Z.

Using this we can generalise the statement from one variable to many variables.

Fist, note that the ideal $\langle \mathbf{a} \rangle$ generated by a non-constant polynomial $\mathbf{a}$ is a special case of an ideal I in $Z[X_1, \ldots, X_n]$ such that $I \cap Z = 0$. In turn, this is equivalent to the condition that Z is a subring of the quotient ring $R = Z[X_1, \ldots, X_n]/I$. Note that the latter is precisely a finitely generated ring over Z.

Noether normalisation theorem. Given the domain Z which is a subring of R which is finitely generated over Z. There is a non-zero element c of Z such that R_c has the following structure:

- R_c contains elements $u_1, \ldots, u_d$ such that the resulting homomorphism $Z_c[T_1, \ldots, T_d] \to R_c$ is one-to-one. Let $S = Z[u_1, \ldots, u_c]$ be the resulting subring of R.
- R contains elements $v_1, \ldots, v_e$ such that $R_c = S_c \cdot v_1 \oplus S_c \cdot v_2 \oplus \cdots \oplus S_c \cdot v_e$; in other words, elements of R_c are S_c linear combinations of the elements $v_1, \ldots, v_e$ which ware linearly independent over S_c.

It follows that multiplication by v_i on R_c is given by a $e \times e$ matrix M_i with entries in S_c. Furthermore, R_c is identified with the subring $S_c[M_1, \ldots, M_e]$ of the ring $M_e(S_c)$ of $e \times e$ matrices with entries in S_c.

In particular, upto inverting one non-zero element c of Z, we can identify R as the ring generated by a finite collection of matrices with entries in a polynomial ring.

Applications

The above results have numerous applications.

Coefficients form a field. We first consider the case when the domain Z is actually a field F.

In this case, any non-constant polynomial is monic.

Since $F[X]$ is a UFD, any polynomial $\mathbf{a}$ can be written as a product of irreducible polynomials.

Note that linear factors correspond to roots in F and can be ignored. Let $\mathbf{p}$ be a factor of $\mathbf{a}$ which has degree at least 2.

An irreducible polynomial $\mathbf{p}$ in $F[X]$ is prime. Moreover, by the GCD algorithm, we see that $E = F[X]/\langle p \rangle$ is a field. Further, E is a finite dimensional vector space over F.

The image η of X in E is a root of $\mathbf{p}$, so that $\mathbf{p} = (X - \eta)\mathbf{q}$ in $E[X]$. It follows that $\mathbf{a}$ has at least one more linear factor in $E[X]$ than it does in $F[X]$.

We can repeating the above steps with $\mathbf{a}$ and $E[X]$. Since the number of linear factors of $\mathbf{a}$ is bounded by its degree, we obtain the following result.

> There is a field L containing F which is finite dimensional as a vector space over F such that a given polynomial $\mathbf{a}$ in $F[X]$ splits into linear factors in $L[X]$.

Such an L is called a *splitting field* for $\mathbf{a}$ over F.

Using a basis of L as a vector space over F, we can identify L with a subring of the matrix ring $M_d(F)$ where d is the dimension of L as a vector space over R.

Roots and Eigenvalues. We now examine the case of a monic polynomial $\mathbf{a}$ in $Z[X]$ where Z is a domain.

As above we can think of $Z[X]/\langle \mathbf{a} \rangle$ as the subring $Z[C_{\mathbf{a}}]$ of $M_d(Z)$. If F is the field of fractions of Z, let $L \supset F$ be the splitting field of $\mathbf{a}$.

We see that every root of $\mathbf{a}$ gives a homomorphism $Z[X]/\langle \mathbf{a} \rangle \to L$ which is identity on Z.

Exercise: Check that each such homomorphism gives an eigenvalue for the matrix $C_{\mathbf{a}}$ as an element of $M_d(L)$.

Exercise: Conversely, if E is a field containing F such that $C_{\mathbf{a}}$ such that we have an eigenvector v for $C_{\mathbf{a}}$ in $M_d(E)$, then show that the associated eigenvalue is a root of $\mathbf{a}$.

Commuting matrices. Suppose A and B are commuting matrices in $M_d(Z)$ and $\mathbf{a}$ (respectively $\mathbf{b}$) are monic polynomials in $Z[X]$ satisfied by them (for example, their characteristic polynomials).

Further suppose we are given ring homomorphism Z to a field L such that $\mathbf{a}$ and $\mathbf{b}$ split into linear factors in $L[X]$.

Exercise: Show that (the images of) A and B in $M_d(L)$ have a common eigenvector. (Hint: Given an eigenvalue α of A in $M_d(L)$ show that the subspace W_α of eigenvectors associated to this eigenvalue satisfies $B(W_\alpha) \subset W_\alpha$. Now take an eigenvector for $B_{|W_\alpha}$.)

Clearly, this cane extended to any finite set of commuting matrices.

General system of equations. As seen above, given a collection S of equations in $\mathbb{Z}[X_1, \ldots, X_n]$ we have a finitely generated ring $R = \mathbb{Z}[X_1, \ldots, X_n]/I$ associated with $S/$

As seen at the start, either $\mathbb{Z} \subset R$ or there is some positive integer n such that $n \in I$.

In case $1 \in I$, the system of equations is *inconsistent* since solving them requires putting $1 = 0$! So we may assume that $n > 1$. In this case, we take a prime factor p of n and consider the image J of I in $\mathbb{F}_p[X_1, \ldots, X_n]$ and study the ring $R = \mathbb{F}_p[X_1, \ldots, X_n]/J$.

Thus, we can formulate the problem of solving the system S to the case where either $\mathbb{Z} \subset R$ or $\mathbb{F}_p \subset R$. The Noether Normalisation theorem thus becomes applicable with Z either $\mathbb{Z}$ or $\mathbb{F}_p$.

We follow notation as in the theorem above.

We let $Z_c \to F$ be a ring homomorphism from Z_c to a field F and $\eta_1, \ldots, \eta_d$ be elements of F. This gives a homomorphism $Z_c[T_1, \ldots, T_d] \to F$ and since $Z_c[T_1, \ldots, T_d] \to S = Z_c[u_1, \ldots, u_d]$ is an isomorphism, we get a homomorphism $S \to F$. This homomorphism takes the entries of the matrices M_i to matrices A_i in $M_e(F)$. Since the matrices M_i commute, so do the matrices A_i. It follows that we can find a simultaneous eigenvector for these matrices in some splitting field L containing F. As seen above, this gives a homomorphism $R \to E$ which solves the system S.

In summary, solving the system of equations S can be reduced to finding simultaneous eigenvectors for a system of commuting matrices. Moreover, this can be done in a field E which is a finite extension of a field F for which we have a homomorphism $Z_c \to F$.

When $Z_c = \mathbb{Z}_c$ for some non-zero integer c, we can take $F = \mathbb{Q}$, so that E is a finite extension of the field of rational numbers.

In both of the above cases, we can take $F = \mathbb{F}_p$ for some prime p, so that E is a finite extension of $\mathbb{F}_p$.

The latter case points to the need to study finite extensions of $\mathbb{F}_p$ which is the topic we take up next!